Data Analytics and Statistical Methods (CS-810F): Project Part 3

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1 Statistical Tests

Our null and alternative hypotheses are as follows:

Null Hypothesis: Weather conditions *(mild, moderate, and severe)* have no effect on students' *willingness to attend in-person classes*.

Alternative Hypothesis: Weather conditions *(mild, moderate, and severe) do have an effect* on students' *willingness to attend in-person classes*.

We used a chi-square test of independence to evaluate whether weather conditions (mild (light rain, temperatures 50–70°F, light winds), moderate (heavy rain, temperatures 30–40°F or 80–90°F, moderate winds), and severe (snow/ice storms, temperatures below 20°F or above 95°F, severe storms, flooding warnings)) were associated with students' willingness to attend in-person classes. This test was appropriate because both variables were categorical. We treated each weather condition as a category, and willingness was measured on an ordinal 1–5 scale representing levels of willingness to attend in person. A rating of 1 indicated not at all willing to attend in person (strong preference for online), 2 indicated slightly unwilling, 3 reflected a neutral stance, 4 indicated willing to attend in person, and 5 indicated very

willing (strong preference for in-person attendance).

2 Descriptive Statistics

The median willingness ratings were 3 for mild weather, 2 for moderate weather, and 1 for severe weather. This means that the median willingness ratings decreased as weather conditions became more severe. Additionally, the IQRs of the ratings for mild, moderate, and severe weather conditions are 2, 2, and 1, respectively. This indicates that willingness has greater variability during mild and moderate weather conditions.

3 Data Visualization

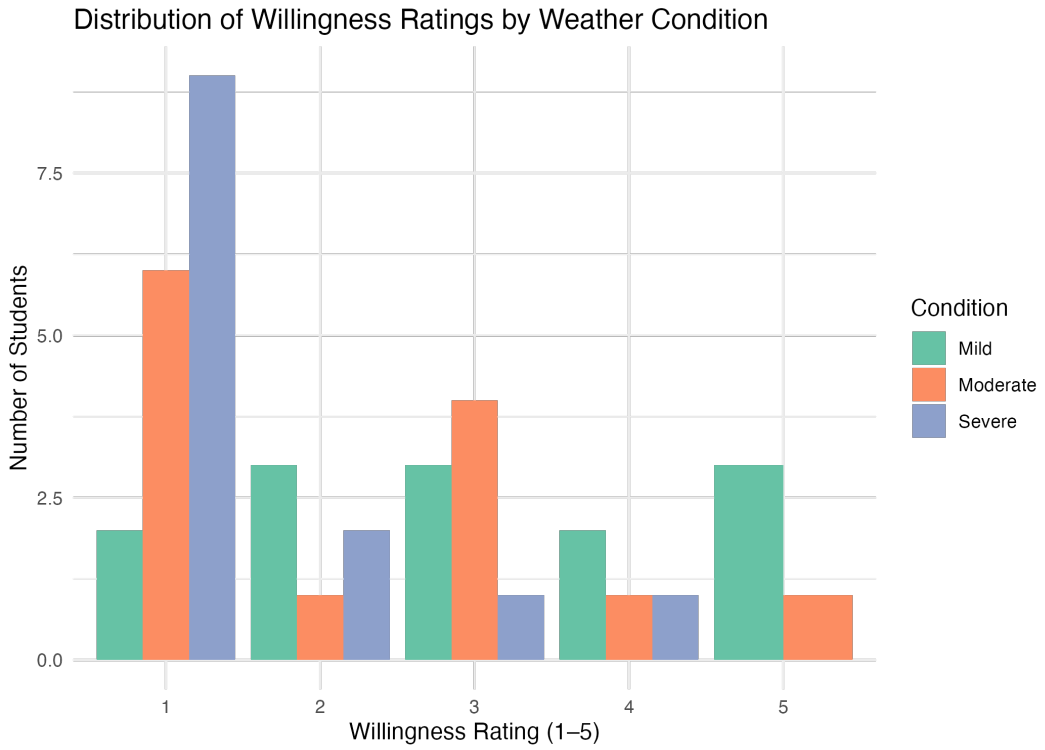


Figure 1: Distribution of willingness ratings (1–5) across mild, moderate, and severe weather conditions.

From both the bar chart and the contingency table, we can see that willingness to attend

Table 1: Contingency table of willingness ratings (1–5) under mild, moderate, and severe weather conditions.

Weather Condition	**1**	**2**	**3**	**4**	**5**
Mild	2	3	3	2	3
Moderate	6	1	4	1	1
Severe	9	2	1	1	0

in-person classes decreases as weather conditions become more severe. The counts are more evenly distributed across the rating scale under mild conditions; however, they shift strongly toward lower willingness ratings under moderate and especially severe weather. Also, severe weather shows the highest concentration of ratings at 1, indicating clear reluctance to attend in person. Lastly, the bar chart depicts the shrinking height of higher-willingness bars as conditions move from mild to moderate to severe. This illustrates a consistent decline in in-person attendance willingness as weather becomes worse.

4 Conclusion of the Test

The chi-square test of independence produced a test statistic of $X^2(8) = 11.103$ with a p-value of 0.1959. Using a significance level of $\alpha = 0.05$, we fail to reject the null hypothesis that weather conditions (mild, moderate, and severe) have no effect on students' *willingness to attend in-person classes*. Although the descriptive statistics and visualizations suggest a downward trend in willingness as weather becomes more severe, this pattern was not statistically significant in our sample. Thus, the "myth" that weather conditions influence students' willingness to attend in person is not confirmed based on this dataset.

5 Effect Size

Although we did not obtain a statistically significant result, we computed Cramer's V as a descriptive measure of effect size. Its value is approximately 0.377, which means a small-to-moderate association between weather conditions and willingness to attend in-person

classes. Moreover, this effect size suggests that, even though the chi-square test did not reach statistical significance, there is still a meaningful pattern in the data consistent with the trends observed in the visualizations.

6 Power Analysis

We conducted a power analysis using the `pwr.chisq.test()` function with Cramer's V as the effect size. This indicates that the statistical power of our test was only about 12.4%, far below the commonly accepted threshold of 80%. Because of this extremely low power, our non-significant result should be interpreted with caution. We will discuss the issues behind such low power in the next section.

7 Weaknesses of the Analysis

There are some weaknesses and limitations of our analysis. First of all, our sample size was very small, which greatly reduced the statistical power of the chi-square test and made it difficult to detect meaningful differences across weather conditions. This was the most important limitation, as many cells in the contingency table had very low expected counts. Secondly, the willingness ratings were measured on an ordinal scale, but the chi-square test treats these values as nominal categories. This limits the amount of information captured about the direction and intensity of students' willingness. Also, we initially cleaned the dataset based on a different analysis plan. However, we later revised our approach and added back variables needed for the chi-square test. Finally, the data relied on subjective self-reported perceptions of weather conditions rather than observed attendance behavior. Thus, the results should be interpreted with caution, as these limitations may have influenced both the statistical test and the conclusions drawn from it.